

*E-Wallet*

**Database Design Document**  
**V 3.0**

**By**

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*Huzaira Sultan*

*NUM-BSCS-2023-07*



**Department of Computer Sciences**  
**Namal University Mianwali,**  
**Pakistan**

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**REVISION HISTORY**

| <b>Date</b> | <b>Version</b> | <b>Description</b>                                                                                                                                                                                                     | <b>Approved by</b> |
|-------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 02/6/24     | V.2.0          | Specialized sub-entities for different transaction types (e.g., CNIC_Transfer, Bill_payment) were introduced to represent specific operations more accurately. Relationships were refined for clarity and correctness. | Mam Asiya          |
| 22/04/24    | V 1.0          | Transaction-related relationships were reviewed; the need for clearer structure using sub-entities for different transaction types was identified.                                                                     | Mam Asiya          |

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# 1. PROJECT OVERVIEW

## 1.1. INTRODUCTION:

An **e-wallet (electronic wallet)** is a digital system that securely stores users' payment information and passwords, allowing them to perform financial transactions electronically. It serves as a virtual bank account, enabling users to send or receive money, pay bills, recharge mobile credit, and more—without the need for physical cash or cards.

The E-Wallet database project has been created to enable an electronic financial system where individuals can manage money electronically. It provides functionalities for user registration, login, maintaining a wallet balance, transferring funds, paying bills, subscribing to mobile packages, and tracking transaction history. The goal is to simulate a secure and efficient financial system using a well-structured relational database.

## 1.2. PROBLEM STATEMENT:

In the modern digital era, users demand quick, safe, and convenient payment methods. Traditional cash-based systems are often slow, insecure, and inaccessible to many. Existing banking systems may not meet the needs of all individuals due to limited access or complexity. The E-Wallet system addresses these issues by providing a digital platform that allows users to perform transactions, transfer funds, and pay for services securely and efficiently.

## 1.3. PROJECT OBJECTIVES:

1. Support wallet balance tracking and multi-currency handling.
2. Allow fund transfers between users, wallets, and CNICs.
3. Enable mobile load.
4. Facilitate utility bill payments (electricity, gas, etc.).
5. Record and maintain transaction history.

## 1.4. DOCUMENT OBJECTIVES:

- Provide a clear overview of the database scope and purpose.
- Describe all entities, attributes, and relationships.
- Present an ERD (Entity Relationship Diagram) and relational schema.
- Explain functional dependencies and normalization process.
- Describe physical table structures, sample data, and query outputs.
- Serve as a blueprint for future development and database administration.

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## 2. DETAILED DATABASE DESIGN

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### 2.1 ENTITY:

| Sr. No | Entity Name           | Description                                                            |
|--------|-----------------------|------------------------------------------------------------------------|
| 01     | User                  | Represents individuals who use the E-Wallet system for transactions.   |
| 02     | Transaction           | Represents any financial action (e.g., transfer, payment) by the user. |
| 03     | Logs                  | Stores login credentials and access logs.                              |
| 04     | Wallet                | Records wallet balance and currency type for each user.                |
| 05     | Bill_Payment          | Represents utility bill payments made by the user.                     |
| 06     | Mobile Load           | Represents mobile credit recharge made by the user.                    |
| 07     | Internal Transfer     | Represents transfers between users using the same wallet type          |
| 08     | Inter_Wallet Transfer | Represents transfers made to users with different wallet types.        |
| 09     | CNIC Transfer         | Represents transfers made using the receiver's CNIC number.            |

### 2.2 DATA DICTIONARY:

#### 1.1 2.2.1 User

| Sr. No | Name        | Data Type    | Constraint  | Description             |
|--------|-------------|--------------|-------------|-------------------------|
| 01     | CNIC_user   | Varchar (15) | Primary key | Unique ID of the user   |
| 02     | Name        | Varchar (50) | Not Null    | Full Name of the User   |
| 03     | Father Name | Varchar (50) | Not Null    | Father Name of the User |
| 04     | Email       | Varchar (50) | Unique      | User's email address    |
| 05     | Phone-no    | Varchar (15) | Unique      | User's mobile number    |
| 06     | Password    | Varchar (25) | Not Null    | User's login password   |
| 07     | Gender      | Varchar (10) | Not Null    | User's Gender           |
| 08     | Province    | Varchar (30) | Not Null    | User's Province         |
| 09     | District    | Varchar (30) | Not Null    | User's District         |
| 10     | Tehsil      | Varchar (30) | Not Null    | User's Tehsil           |

#### 1.2 2.2.2 Transaction

| Sr. No | Name           | Data Type      | Constraint  | Description                      |
|--------|----------------|----------------|-------------|----------------------------------|
| 01     | Transcation-ID | Int            | Primary Key | Unique transaction identifier    |
| 02     | CNIC User      | Varchar (15)   | Foreign Key | Refers to User                   |
| 03     | Amount         | Decimal (10,2) | Not Null    | Amount transferred               |
| 04     | Timestamp      | DateTime       | Not Null    | Date and time of the transaction |

|    |                 |          |          |                               |
|----|-----------------|----------|----------|-------------------------------|
| 05 | TransactionType | Char (2) | Not Null | Type of transaction (S, D, C) |
|----|-----------------|----------|----------|-------------------------------|

### 2.2.3 Logs

| Sr. No | Name         | Data Type    | Constraint  | Description                              |
|--------|--------------|--------------|-------------|------------------------------------------|
| 01     | Log_id       | Int          | Primary Key | Unique Login ID of the user              |
| 02     | CNIC-User    | Varchar (15) | Foreign Key | Refers to User                           |
| 03     | Login_Time   | DateTime     | Not Null    | Records the exact date and time of login |
| 04     | Log_out_time | DateTime     | Not Null    | Records the log out time                 |

### 1.3 2.2.4 Wallet

| Sr No | Name          | Data Type     | Constraint                     | Description             |
|-------|---------------|---------------|--------------------------------|-------------------------|
| 1     | Wallet_id     | INT           | PK, NOT NULL, UNIQUE           | Wallet ID               |
| 2     | CNIC_user     | VARCHAR(50)   | FK → User(CNIC_user), NOT NULL | Owner CNIC              |
| 3     | Balance       | DECIMAL(10,2) | DEFAULT 0.0                    | Wallet balance          |
| 4     | Creation_date | DATE          | NOT NULL                       | Date of wallet creation |
| 5     | Wallet_type   | VARCHAR(20)   | CHECK (value in enum)          | Wallet type             |
| 6     | Wallet_pin    | VARCHAR(10)   | NOT NULL                       | Wallet PIN              |

### 1.4 2.2.5 Bill Payment

| Sr No | Name             | Data Type   | Constraint                           | Description                       |
|-------|------------------|-------------|--------------------------------------|-----------------------------------|
| 1     | B_Transaction_id | INT         | PK, FK → Transaction(Transaction_id) | Linked transaction ID             |
| 2     | BillType         | VARCHAR(50) | NOT NULL                             | Type of bill (e.g. gas, electric) |
| 3     | Reference_no     | VARCHAR(50) | NOT NULL                             | Reference number of the bill      |
| 4     | Customer_id      | VARCHAR(50) | NOT NULL                             | Customer ID of the bill           |

### 1.5 2.2.7 Mobile Load

| Sr No | Name             | Data Type   | Constraint                           | Description            |
|-------|------------------|-------------|--------------------------------------|------------------------|
| 1     | M_Transaction_id | INT         | PK, FK → Transaction(Transaction_id) | Linked transaction ID  |
| 2     | Mobile_number    | VARCHAR(15) | CHECK (valid format), NOT NULL       | Number to be recharged |

### 2.2.8 Internal Transfer:

| Sr No | Name              | Data Type   | Constraint                           | Description                       |
|-------|-------------------|-------------|--------------------------------------|-----------------------------------|
| 1     | IN_Transaction_id | INT         | PK, FK → Transaction(Transaction_id) | Linked transaction ID             |
| 2     | Recipient_Phone   | VARCHAR(15) | FK → User(Phone_num)                 | Internal recipient's phone number |

### 2.2.9 Inter Wallet Transfer:

| Sr No | Name                  | Data Type   | Constraint                           | Description              |
|-------|-----------------------|-------------|--------------------------------------|--------------------------|
| 1     | I_Transaction_id      | INT         | PK, FK → Transaction(Transaction_id) | Linked transaction ID    |
| 2     | Recipient_Phone       | VARCHAR(15) | FK → User(Phone_num)                 | Recipient's phone number |
| 3     | Recipient_Wallet_Type | VARCHAR(20) | CHECK (value in enum)                | Receiver's wallet type   |

### 2.2.10. CNIC Transfer:

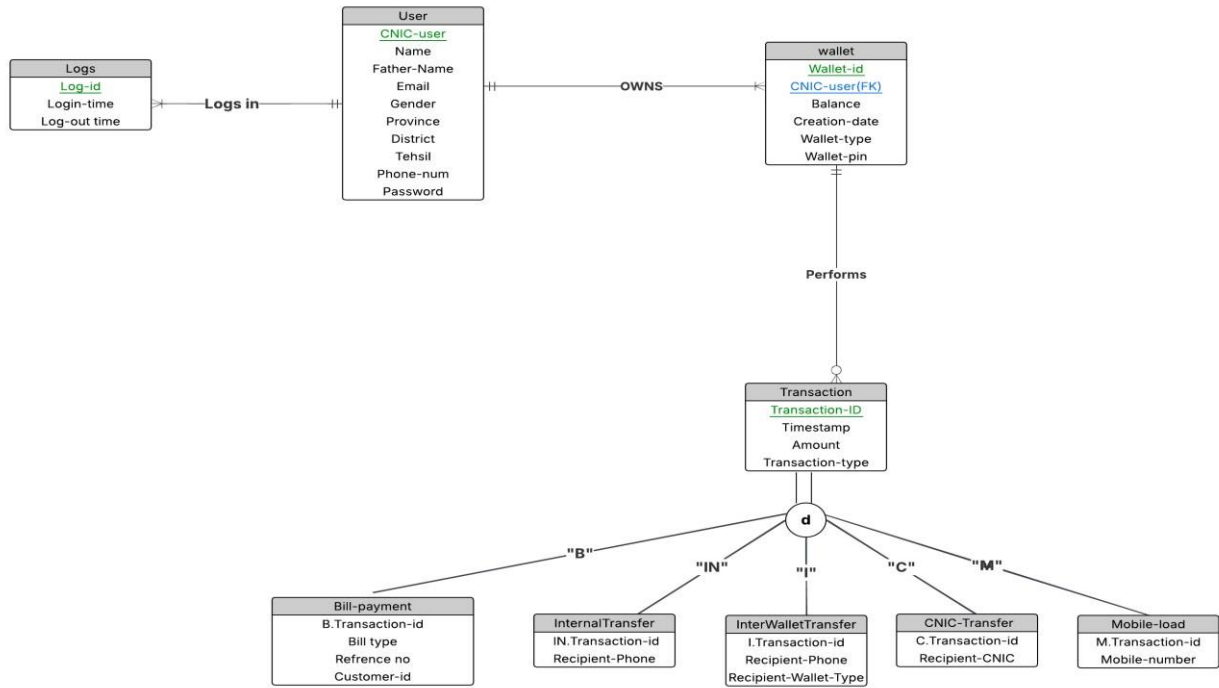
| Sr No | Name             | Data Type   | Constraint                           | Description           |
|-------|------------------|-------------|--------------------------------------|-----------------------|
| 1     | C_Transaction_id | INT         | PK, FK → Transaction(Transaction_id) | Linked transaction ID |
| 2     | Recipient_CNIC   | VARCHAR(15) | FK → User(CNIC_user)                 | Receiver's CNIC       |

## 2.3 RELATIONSHIPS:

| Sr. No | Participating Entities            | Relationship | Business Rule                                                             |
|--------|-----------------------------------|--------------|---------------------------------------------------------------------------|
| 1      | User – Wallet                     | Owns         | One user <b>owns</b> one or more wallets.                                 |
| 2      | User – Logs                       | Logs in      | One user can <b>log in</b> multiple times, creating multiple log records. |
| 3      | Wallet – Transaction              | Performs     | One wallet <b>performs</b> one or more transactions.                      |
| 4      | Transaction – Bill_payment        | "B"          | A bill payment is a <b>type</b> of transaction.                           |
| 5      | Transaction – InternalTransfer    | "IN"         | An internal transfer is a <b>type</b> of transaction.                     |
| 6      | Transaction – InterWalletTransfer | "I"          | An inter-wallet transfer is a <b>type</b> of transaction.                 |
| 7      | Transaction – CNIC_Transfer       | "C"          | A CNIC transfer is a <b>type</b> of transaction.                          |
| 8      | Transaction – Mobile_load         | "M"          | A mobile load is a <b>type</b> of transaction.                            |



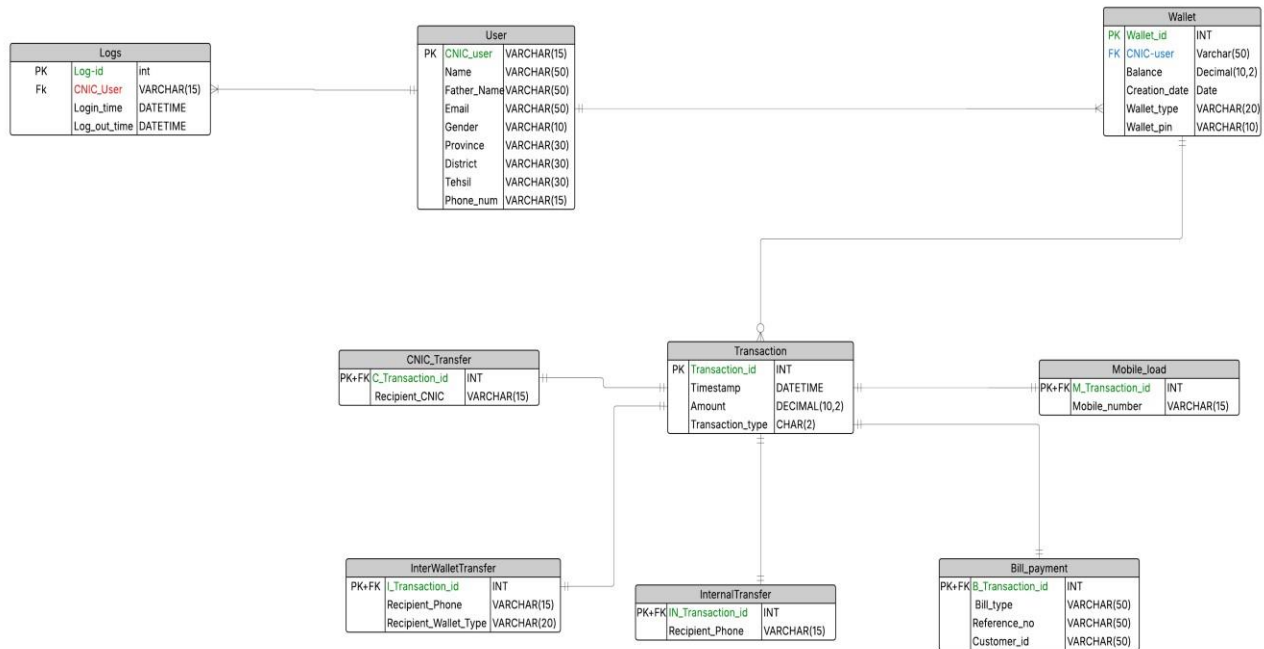
## 2.4 ENTITY RELATIONSHIP DIAGRAM: [1]



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## 3: LOGICAL DATABASE DESIGN

### 3.1. RELATIONAL SCHEMA: [2] [3]



### 3.2. FUNCTIONAL DEPENDENCIES:

- **User**  
 $CNIC\_user \rightarrow Name, Father\_Name, Email, Gender, Province, District, Tehsil, Phone\_num$
- **Wallet**  
 $Wallet\_id \rightarrow CNIC\_user, Balance, Creation\_date, Wallet\_type, Wallet\_pin$
- **Logs**  
 $Log\_id \rightarrow CNIC\_user, Login\_time, Log\_out\_time$
- **Transaction**  
 $Transaction\_id \rightarrow Timestamp, Amount, Transaction\_type$

- **CNIC\_Transfer**  
C\_Transaction\_id → Recipient\_CNIC
- **InterWalletTransfer**  
I\_Transaction\_id → Recipient\_Phone, Recipient\_Wallet\_Type
- **InternalTransfer**  
IN\_Transaction\_id → Recipient\_Phone
- **Bill\_payment**  
B\_Transaction\_id → BillType, Reference\_no, Customer\_id
- **Mobile\_load**  
M\_Transaction\_id → Mobile\_number

### 3.3. NORMALIZATION:

Normalization is the process of organizing data to reduce redundancy and improve data integrity. It is applied step by step through normal forms. For the E-Wallet system, the relations were passed through First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF) to make sure that the structure is clean and efficient.

#### 1.6 First Normal Form (1NF)

- All tables have atomic values and unique rows.
- Repeating groups are eliminated.

#### 1.7 Second Normal Form (2NF)

- All non-key attributes are fully functionally dependent on the primary key.
- Composite keys are avoided (except where necessary in bridge tables).

#### 1.8 Third Normal Form (3NF)

- All transitive dependencies are removed.
- Each non-key attribute depends only on the key.

#### Final 3NF Relations:

1. **User**(CNIC\_user, Name, Father\_Name, Email, Gender, Province, District, Tehsil, Phone\_num)

2. **Wallet**(Wallet\_id, CNIC\_user, Balance, Creation\_date, Wallet\_type, Wallet\_pin)
3. **Logs**(Log\_id, CNIC\_User, Login\_time, Log\_out\_time)
4. **Transaction**(Transaction\_id, Timestamp, Amount, Transaction\_type)
5. **CNIC\_Transfer**(C\_Transaction\_id, Recipient\_CNIC)
6. **Mobile\_load**(M\_Transaction\_id, Mobile\_number)
7. **InterWalletTransfer**(I\_Transaction\_id, Recipient\_Phone, Recipient\_Wallet\_Type)
8. **InternalTransfer**(IN\_Transaction\_id, Recipient\_Phone)
9. **Bill\_payment**(B\_Transaction\_id, Bill\_type, Reference\_no, Customer\_id)# **Conclusion**

After applying normalization:

- All the relations are in **Third Normal Form (3NF)**.
- The structure now avoids redundancy, ensures consistency, and maintains data integrity.

### 1.9 Final Normalized Relations in 3NF

| Table Name            | Primary Key    | Normal Form |
|-----------------------|----------------|-------------|
| User                  | CNIC_User      | 3NF         |
| Logs                  | Phone_No       | 3NF         |
| Wallet                | CNIC_User      | 3NF         |
| Transaction           | Transaction_ID | 3NF         |
| Internal_Transfer     | Transaction_ID | 3NF         |
| Inter_Wallet_Transfer | Transaction_ID | 3NF         |
| CNIC_Transfer         | Transaction_ID | 3NF         |
| Bill Payment          | Bill_ID        | 3NF         |
| Mobile_Load           | Load_ID        | 3NF         |

## 4: PHYSICAL DATABASE DESIGN

## 4.1. STRUCTURE OF THE TABLES:

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Table 1</b> | <b>User</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Query</b>   | DESC User;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Output</b>  | <pre>mysql&gt; DESC User; +-----+-----+-----+-----+-----+-----+   Field        Type            Null   Key   Default   Extra   +-----+-----+-----+-----+-----+-----+   CNIC_user    varchar(18)     NO     PRI   NULL                Name         varchar(50)     YES          NULL                Father_Name   varchar(50)     YES          NULL                Email        varchar(50)     YES          NULL                Gender       varchar(10)     YES          NULL                Province     varchar(30)     YES          NULL                District     varchar(30)     YES          NULL                Tehsil       varchar(30)     YES          NULL                Phone_num    varchar(15)     YES          NULL              +-----+-----+-----+-----+-----+-----+ 9 rows in set (0.02 sec)</pre> <p>Fig 4.1.1</p> |
| <b>Table 2</b> | <b>Wallet</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Query</b>   | DESC Wallet;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Output</b>  | <pre>mysql&gt; DESC wallet; +-----+-----+-----+-----+-----+-----+   Field        Type            Null   Key   Default   Extra   +-----+-----+-----+-----+-----+-----+   Wallet_id    int             NO     PRI   NULL                CNIC_user    varchar(18)     YES    MUL   NULL                Balance      decimal(10,2)   YES    MUL   NULL                Creation_date   date           YES          NULL                Wallet_type   varchar(20)     YES          NULL                Wallet_pin   varchar(10)     YES          NULL              +-----+-----+-----+-----+-----+-----+ 6 rows in set (0.00 sec)</pre> <p>Fig 4.1.2</p>                                                                                                                                                                                       |
| <b>Table 3</b> | <b>Logs</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Query</b>   | DESC logs;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Output</b>  | <pre>mysql&gt; DESC LOGS; +-----+-----+-----+-----+-----+-----+   Field        Type            Null   Key   Default   Extra   +-----+-----+-----+-----+-----+-----+   Log_id       int             NO     PRI   NULL                CNIC_User    varchar(18)     YES    MUL   NULL                Login_time   datetime        YES    MUL   NULL                Log_out_time   datetime       YES          NULL              +-----+-----+-----+-----+-----+-----+ 4 rows in set (0.00 sec)</pre> <p>Fig 4.1.3</p>                                                                                                                                                                                                                                                                                                                       |

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Table 4</b> | <b>Transaction</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Query</b>   | DESC Transaction;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Output</b>  | <pre>mysql&gt; DESC transaction; +-----+-----+-----+-----+-----+-----+   Field            Type            Null   Key   Default   Extra   +-----+-----+-----+-----+-----+-----+   Transaction_id   int             NO     PRI   NULL                Timestamp        datetime        YES    MUL   NULL                Amount           decimal(10,2)   YES    MUL   NULL                Transaction_type   char(2)         YES          NULL                wallet_id        int             YES    MUL   NULL              +-----+-----+-----+-----+-----+-----+ 5 rows in set (0.00 sec)</pre> |
| Fig 4.1.4      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Table 5</b> | <b>CNIC_transfer</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Query</b>   | DESC CNIC_transfer;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Output</b>  | <pre>mysql&gt; DESC CNIC_transfer; +-----+-----+-----+-----+-----+-----+   Field            Type            Null   Key   Default   Extra   +-----+-----+-----+-----+-----+-----+   C_Transaction_id   int             NO     PRI   NULL                Recipient_CNIC    varchar(20)     YES    MUL   NULL              +-----+-----+-----+-----+-----+-----+ 2 rows in set (0.00 sec)</pre>                                                                                                                                                                                                    |
| Fig 4.1.5      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Table 6</b> | <b>Mobile Load</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Query</b>   | DESC mobile_load;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Output</b>  | <pre>mysql&gt; DESC mobile_load; +-----+-----+-----+-----+-----+-----+   Field            Type            Null   Key   Default   Extra   +-----+-----+-----+-----+-----+-----+   M_Transaction_id   int             NO     PRI   NULL                Mobile_number      varchar(20)     YES    MUL   NULL              +-----+-----+-----+-----+-----+-----+ 2 rows in set (0.00 sec)</pre>                                                                                                                                                                                                     |
| Fig 4.1.6      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Table 7</b> | <b>Inter wallet transfer</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Query</b>   | DESC interwallettransfer;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Output</b> | <pre>mysql&gt; DESC interwallettransfer; +-----+-----+-----+-----+-----+-----+   Field            Type        Null   Key   Default   Extra   +-----+-----+-----+-----+-----+-----+   I_Transaction_id   int         NO     PRI   NULL                Recipient_Phone    varchar(20)   YES    MUL   NULL                Recipient_Wallet_Type   varchar(30)   YES          NULL              +-----+-----+-----+-----+-----+-----+ 3 rows in set (0.00 sec)</pre> |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Fig 4.1.7

|                |                          |
|----------------|--------------------------|
| <b>Table 8</b> | <b>Internal Transfer</b> |
|----------------|--------------------------|

|              |                         |
|--------------|-------------------------|
| <b>Query</b> | DESC internal_transfer; |
|--------------|-------------------------|

|               |                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Output</b> | <pre>mysql&gt; DESC internaltransfer; +-----+-----+-----+-----+-----+-----+   Field            Type        Null   Key   Default   Extra   +-----+-----+-----+-----+-----+-----+   IN_Transaction_id   int         NO     PRI   NULL                Recipient_Phone    varchar(20)   YES    MUL   NULL              +-----+-----+-----+-----+-----+-----+ 2 rows in set (0.00 sec)</pre> |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Fig 4.1.8

|                |                     |
|----------------|---------------------|
| <b>Table 9</b> | <b>Bill payment</b> |
|----------------|---------------------|

|              |                    |
|--------------|--------------------|
| <b>Query</b> | DESC bill_payment; |
|--------------|--------------------|

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Output</b> | <pre>mysql&gt; DESC bill_payment; +-----+-----+-----+-----+-----+-----+   Field            Type        Null   Key   Default   Extra   +-----+-----+-----+-----+-----+-----+   B_Transaction_id   int         NO     PRI   NULL                Bill_type         varchar(50)   YES          NULL                Reference_no      varchar(50)   YES    MUL   NULL                Customer_id       varchar(50)   YES          NULL              +-----+-----+-----+-----+-----+-----+ 4 rows in set (0.00 sec)</pre> |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Fig 4.1.9

## 4.2. DATA SAMPLES INSIDE TABLES: [3]

|                |                     |
|----------------|---------------------|
| <b>Table 1</b> | <b>User</b>         |
| <b>Query</b>   | Select * from User; |

| Output           | <pre>mysql&gt; select * from user;</pre> <table><thead><tr><th>CNIC_user</th><th>Name</th><th>Father_Name</th><th>Email</th><th>Gender</th><th>Province</th><th>District</th><th>Tehsil</th><th>Phone_num</th></tr></thead><tbody><tr><td>35202-1234567-1</td><td>Ali</td><td>Ahmed</td><td>ali@gmail.com</td><td>Male</td><td>Punjab</td><td>Lahore</td><td>Model Town</td><td>03000000001</td></tr><tr><td>35202-1234567-10</td><td>Nadia</td><td>Bano</td><td>nadia@gmail.com</td><td>Female</td><td>Punjab</td><td>Gujranwala</td><td>City</td><td>03000000010</td></tr><tr><td>35202-1234567-11</td><td>Rehan</td><td>Shah</td><td>rehan@gmail.com</td><td>Male</td><td>Punjab</td><td>Bahawalpur</td><td>Ahmedpur</td><td>03000000011</td></tr><tr><td>35202-1234567-12</td><td>Tania</td><td>Zafar</td><td>tania@gmail.com</td><td>Female</td><td>Sindh</td><td>Sukkur</td><td>Old City</td><td>03000000012</td></tr><tr><td>35202-1234567-13</td><td>Imran</td><td>Qureshi</td><td>imran@gmail.com</td><td>Male</td><td>Punjab</td><td>Sialkot</td><td>Daska</td><td>03000000013</td></tr><tr><td>35202-1234567-14</td><td>Zara</td><td>Latif</td><td>zara@gmail.com</td><td>Female</td><td>KPK</td><td>Abbottabad</td><td>Main</td><td>03000000014</td></tr><tr><td>35202-1234567-15</td><td>Kamran</td><td>Farooq</td><td>kamran@gmail.com</td><td>Male</td><td>Punjab</td><td>Okara</td><td>Renala</td><td>03000000015</td></tr><tr><td>35202-1234567-2</td><td>Sara</td><td>Khan</td><td>sara@gmail.com</td><td>Female</td><td>Sindh</td><td>Karachi</td><td>Gulshan</td><td>03000000002</td></tr><tr><td>35202-1234567-3</td><td>Usman</td><td>Ali</td><td>usman@gmail.com</td><td>Male</td><td>Punjab</td><td>Multan</td><td>Shujabad</td><td>03000000003</td></tr><tr><td>35202-1234567-4</td><td>Ayesha</td><td>Noor</td><td>ayesha@gmail.com</td><td>Female</td><td>Punjab</td><td>Lahore</td><td>Iqbal Town</td><td>03000000004</td></tr><tr><td>35202-1234567-5</td><td>Bilal</td><td>Zahid</td><td>bilal@gmail.com</td><td>Male</td><td>KPK</td><td>Peshawar</td><td>Town 1</td><td>03000000005</td></tr><tr><td>35202-1234567-6</td><td>Hina</td><td>Iqbal</td><td>hina@gmail.com</td><td>Female</td><td>Balochistan</td><td>Quetta</td><td>Sariab</td><td>03000000006</td></tr><tr><td>35202-1234567-7</td><td>Hamza</td><td>Kamal</td><td>hamza@gmail.com</td><td>Male</td><td>Punjab</td><td>Faisalabad</td><td>Lyallpur</td><td>03000000007</td></tr><tr><td>35202-1234567-8</td><td>Farah</td><td>Saeed</td><td>farah@gmail.com</td><td>Female</td><td>Sindh</td><td>Hyderabad</td><td>Latifabad</td><td>03000000008</td></tr><tr><td>35202-1234567-9</td><td>Danish</td><td>Ali</td><td>danish@gmail.com</td><td>Male</td><td>Punjab</td><td>Rawalpindi</td><td>Saddar</td><td>03000000009</td></tr></tbody></table> <p>15 rows in set (0.00 sec)</p> | CNIC_user           | Name                | Father_Name | Email         | Gender      | Province        | District            | Tehsil              | Phone_num | 35202-1234567-1 | Ali                 | Ahmed               | ali@gmail.com | Male            | Punjab              | Lahore              | Model Town | 03000000001     | 35202-1234567-10    | Nadia               | Bano    | nadia@gmail.com | Female              | Punjab              | Gujranwala | City            | 03000000010         | 35202-1234567-11    | Rehan    | Shah            | rehan@gmail.com     | Male                | Punjab  | Bahawalpur      | Ahmedpur            | 03000000011         | 35202-1234567-12 | Tania           | Zafar               | tania@gmail.com     | Female   | Sindh            | Sukkur              | Old City            | 03000000012 | 35202-1234567-13 | Imran               | Qureshi             | imran@gmail.com | Male             | Punjab              | Sialkot             | Daska    | 03000000013      | 35202-1234567-14    | Zara                | Latif   | zara@gmail.com   | Female              | KPK                 | Abbottabad | Main             | 03000000014         | 35202-1234567-15    | Kamran   | Farooq | kamran@gmail.com | Male             | Punjab  | Okara      | Renala     | 03000000015 | 35202-1234567-2 | Sara             | Khan    | sara@gmail.com | Female   | Sindh | Karachi | Gulshan          | 03000000002 | 35202-1234567-3 | Usman      | Ali  | usman@gmail.com | Male             | Punjab  | Multan     | Shujabad | 03000000003 | 35202-1234567-4 | Ayesha           | Noor    | ayesha@gmail.com | Female     | Punjab | Lahore | Iqbal Town | 03000000004 | 35202-1234567-5 | Bilal | Zahid | bilal@gmail.com | Male | KPK | Peshawar | Town 1 | 03000000005 | 35202-1234567-6 | Hina | Iqbal | hina@gmail.com | Female | Balochistan | Quetta | Sariab | 03000000006 | 35202-1234567-7 | Hamza | Kamal | hamza@gmail.com | Male | Punjab | Faisalabad | Lyallpur | 03000000007 | 35202-1234567-8 | Farah | Saeed | farah@gmail.com | Female | Sindh | Hyderabad | Latifabad | 03000000008 | 35202-1234567-9 | Danish | Ali | danish@gmail.com | Male | Punjab | Rawalpindi | Saddar | 03000000009 |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------|-------------|---------------|-------------|-----------------|---------------------|---------------------|-----------|-----------------|---------------------|---------------------|---------------|-----------------|---------------------|---------------------|------------|-----------------|---------------------|---------------------|---------|-----------------|---------------------|---------------------|------------|-----------------|---------------------|---------------------|----------|-----------------|---------------------|---------------------|---------|-----------------|---------------------|---------------------|------------------|-----------------|---------------------|---------------------|----------|------------------|---------------------|---------------------|-------------|------------------|---------------------|---------------------|-----------------|------------------|---------------------|---------------------|----------|------------------|---------------------|---------------------|---------|------------------|---------------------|---------------------|------------|------------------|---------------------|---------------------|----------|--------|------------------|------------------|---------|------------|------------|-------------|-----------------|------------------|---------|----------------|----------|-------|---------|------------------|-------------|-----------------|------------|------|-----------------|------------------|---------|------------|----------|-------------|-----------------|------------------|---------|------------------|------------|--------|--------|------------|-------------|-----------------|-------|-------|-----------------|------|-----|----------|--------|-------------|-----------------|------|-------|----------------|--------|-------------|--------|--------|-------------|-----------------|-------|-------|-----------------|------|--------|------------|----------|-------------|-----------------|-------|-------|-----------------|--------|-------|-----------|-----------|-------------|-----------------|--------|-----|------------------|------|--------|------------|--------|-------------|
| CNIC_user        | Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Father_Name         | Email               | Gender      | Province      | District    | Tehsil          | Phone_num           |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 35202-1234567-1  | Ali                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Ahmed               | ali@gmail.com       | Male        | Punjab        | Lahore      | Model Town      | 03000000001         |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 35202-1234567-10 | Nadia                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Bano                | nadia@gmail.com     | Female      | Punjab        | Gujranwala  | City            | 03000000010         |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 35202-1234567-11 | Rehan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Shah                | rehan@gmail.com     | Male        | Punjab        | Bahawalpur  | Ahmedpur        | 03000000011         |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 35202-1234567-12 | Tania                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Zafar               | tania@gmail.com     | Female      | Sindh         | Sukkur      | Old City        | 03000000012         |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 35202-1234567-13 | Imran                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Qureshi             | imran@gmail.com     | Male        | Punjab        | Sialkot     | Daska           | 03000000013         |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 35202-1234567-14 | Zara                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Latif               | zara@gmail.com      | Female      | KPK           | Abbottabad  | Main            | 03000000014         |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 35202-1234567-15 | Kamran                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Farooq              | kamran@gmail.com    | Male        | Punjab        | Okara       | Renala          | 03000000015         |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 35202-1234567-2  | Sara                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Khan                | sara@gmail.com      | Female      | Sindh         | Karachi     | Gulshan         | 03000000002         |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 35202-1234567-3  | Usman                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Ali                 | usman@gmail.com     | Male        | Punjab        | Multan      | Shujabad        | 03000000003         |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 35202-1234567-4  | Ayesha                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Noor                | ayesha@gmail.com    | Female      | Punjab        | Lahore      | Iqbal Town      | 03000000004         |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 35202-1234567-5  | Bilal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Zahid               | bilal@gmail.com     | Male        | KPK           | Peshawar    | Town 1          | 03000000005         |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 35202-1234567-6  | Hina                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Iqbal               | hina@gmail.com      | Female      | Balochistan   | Quetta      | Sariab          | 03000000006         |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 35202-1234567-7  | Hamza                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Kamal               | hamza@gmail.com     | Male        | Punjab        | Faisalabad  | Lyallpur        | 03000000007         |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 35202-1234567-8  | Farah                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Saeed               | farah@gmail.com     | Female      | Sindh         | Hyderabad   | Latifabad       | 03000000008         |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 35202-1234567-9  | Danish                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Ali                 | danish@gmail.com    | Male        | Punjab        | Rawalpindi  | Saddar          | 03000000009         |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| Fig 4.2.1        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                     |                     |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| Table 2          | Wallet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                     |                     |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| Query            | Select * from Wallet;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                     |                     |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| Output           | <pre>mysql&gt; select * from wallet;</pre> <table><thead><tr><th>Wallet_id</th><th>CNIC_user</th><th>Balance</th><th>Creation_date</th><th>Wallet_type</th><th>Wallet_pin</th></tr></thead><tbody><tr><td>1</td><td>35202-1234567-1</td><td>9500.00</td><td>2023-01-01</td><td>EasyPaisha</td><td>1234</td></tr><tr><td>2</td><td>35202-1234567-2</td><td>8700.00</td><td>2023-01-02</td><td>JazzCash</td><td>2345</td></tr><tr><td>3</td><td>35202-1234567-3</td><td>7800.00</td><td>2023-01-03</td><td>EasyPaisha</td><td>3456</td></tr><tr><td>4</td><td>35202-1234567-4</td><td>6600.00</td><td>2023-01-04</td><td>JazzCash</td><td>4567</td></tr><tr><td>5</td><td>35202-1234567-5</td><td>5400.00</td><td>2023-01-05</td><td>EasyPaisha</td><td>5678</td></tr><tr><td>6</td><td>35202-1234567-6</td><td>4300.00</td><td>2023-01-06</td><td>JazzCash</td><td>6789</td></tr><tr><td>7</td><td>35202-1234567-7</td><td>3900.00</td><td>2023-01-07</td><td>EasyPaisha</td><td>7890</td></tr><tr><td>8</td><td>35202-1234567-8</td><td>2850.00</td><td>2023-01-08</td><td>JazzCash</td><td>8901</td></tr><tr><td>9</td><td>35202-1234567-9</td><td>1750.00</td><td>2023-01-09</td><td>EasyPaisha</td><td>9012</td></tr><tr><td>10</td><td>35202-1234567-10</td><td>650.00</td><td>2023-01-10</td><td>JazzCash</td><td>0123</td></tr><tr><td>11</td><td>35202-1234567-11</td><td>9050.00</td><td>2023-01-11</td><td>EasyPaisha</td><td>1470</td></tr><tr><td>12</td><td>35202-1234567-12</td><td>7950.00</td><td>2023-01-12</td><td>JazzCash</td><td>2581</td></tr><tr><td>13</td><td>35202-1234567-13</td><td>6850.00</td><td>2023-01-13</td><td>EasyPaisha</td><td>3692</td></tr><tr><td>14</td><td>35202-1234567-14</td><td>5750.00</td><td>2023-01-14</td><td>JazzCash</td><td>4713</td></tr><tr><td>15</td><td>35202-1234567-15</td><td>4650.00</td><td>2023-01-15</td><td>EasyPaisha</td><td>5824</td></tr></tbody></table> <p>15 rows in set (0.00 sec)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Wallet_id           | CNIC_user           | Balance     | Creation_date | Wallet_type | Wallet_pin      | 1                   | 35202-1234567-1     | 9500.00   | 2023-01-01      | EasyPaisha          | 1234                | 2             | 35202-1234567-2 | 8700.00             | 2023-01-02          | JazzCash   | 2345            | 3                   | 35202-1234567-3     | 7800.00 | 2023-01-03      | EasyPaisha          | 3456                | 4          | 35202-1234567-4 | 6600.00             | 2023-01-04          | JazzCash | 4567            | 5                   | 35202-1234567-5     | 5400.00 | 2023-01-05      | EasyPaisha          | 5678                | 6                | 35202-1234567-6 | 4300.00             | 2023-01-06          | JazzCash | 6789             | 7                   | 35202-1234567-7     | 3900.00     | 2023-01-07       | EasyPaisha          | 7890                | 8               | 35202-1234567-8  | 2850.00             | 2023-01-08          | JazzCash | 8901             | 9                   | 35202-1234567-9     | 1750.00 | 2023-01-09       | EasyPaisha          | 9012                | 10         | 35202-1234567-10 | 650.00              | 2023-01-10          | JazzCash | 0123   | 11               | 35202-1234567-11 | 9050.00 | 2023-01-11 | EasyPaisha | 1470        | 12              | 35202-1234567-12 | 7950.00 | 2023-01-12     | JazzCash | 2581  | 13      | 35202-1234567-13 | 6850.00     | 2023-01-13      | EasyPaisha | 3692 | 14              | 35202-1234567-14 | 5750.00 | 2023-01-14 | JazzCash | 4713        | 15              | 35202-1234567-15 | 4650.00 | 2023-01-15       | EasyPaisha | 5824   |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| Wallet_id        | CNIC_user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Balance             | Creation_date       | Wallet_type | Wallet_pin    |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 1                | 35202-1234567-1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 9500.00             | 2023-01-01          | EasyPaisha  | 1234          |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 2                | 35202-1234567-2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 8700.00             | 2023-01-02          | JazzCash    | 2345          |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 3                | 35202-1234567-3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7800.00             | 2023-01-03          | EasyPaisha  | 3456          |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 4                | 35202-1234567-4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 6600.00             | 2023-01-04          | JazzCash    | 4567          |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 5                | 35202-1234567-5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 5400.00             | 2023-01-05          | EasyPaisha  | 5678          |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 6                | 35202-1234567-6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4300.00             | 2023-01-06          | JazzCash    | 6789          |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 7                | 35202-1234567-7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3900.00             | 2023-01-07          | EasyPaisha  | 7890          |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 8                | 35202-1234567-8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2850.00             | 2023-01-08          | JazzCash    | 8901          |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 9                | 35202-1234567-9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1750.00             | 2023-01-09          | EasyPaisha  | 9012          |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 10               | 35202-1234567-10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 650.00              | 2023-01-10          | JazzCash    | 0123          |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 11               | 35202-1234567-11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 9050.00             | 2023-01-11          | EasyPaisha  | 1470          |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 12               | 35202-1234567-12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 7950.00             | 2023-01-12          | JazzCash    | 2581          |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 13               | 35202-1234567-13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 6850.00             | 2023-01-13          | EasyPaisha  | 3692          |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 14               | 35202-1234567-14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 5750.00             | 2023-01-14          | JazzCash    | 4713          |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 15               | 35202-1234567-15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4650.00             | 2023-01-15          | EasyPaisha  | 5824          |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| Fig 4.2.2        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                     |                     |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| Table 3          | Logs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                     |                     |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| Query            | Select * from logs;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                     |                     |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| Output           | <pre>mysql&gt; select * from logs;</pre> <table><thead><tr><th>Log_id</th><th>CNIC_User</th><th>Login_time</th><th>Log_out_time</th></tr></thead><tbody><tr><td>1</td><td>35202-1234567-1</td><td>2025-06-02 12:53:13</td><td>2025-06-02 12:53:13</td></tr><tr><td>2</td><td>35202-1234567-2</td><td>2025-06-02 12:53:13</td><td>2025-06-02 12:53:13</td></tr><tr><td>3</td><td>35202-1234567-3</td><td>2025-06-02 12:53:13</td><td>2025-06-02 12:53:13</td></tr><tr><td>4</td><td>35202-1234567-4</td><td>2025-06-02 12:53:13</td><td>2025-06-02 12:53:13</td></tr><tr><td>5</td><td>35202-1234567-5</td><td>2025-06-02 12:53:13</td><td>2025-06-02 12:53:13</td></tr><tr><td>6</td><td>35202-1234567-6</td><td>2025-06-02 12:53:13</td><td>2025-06-02 12:53:13</td></tr><tr><td>7</td><td>35202-1234567-7</td><td>2025-06-02 12:53:13</td><td>2025-06-02 12:53:13</td></tr><tr><td>8</td><td>35202-1234567-8</td><td>2025-06-02 12:53:13</td><td>2025-06-02 12:53:13</td></tr><tr><td>9</td><td>35202-1234567-9</td><td>2025-06-02 12:53:13</td><td>2025-06-02 12:53:13</td></tr><tr><td>10</td><td>35202-1234567-10</td><td>2025-06-02 12:53:13</td><td>2025-06-02 12:53:13</td></tr><tr><td>11</td><td>35202-1234567-11</td><td>2025-06-02 12:53:13</td><td>2025-06-02 12:53:13</td></tr><tr><td>12</td><td>35202-1234567-12</td><td>2025-06-02 12:53:13</td><td>2025-06-02 12:53:13</td></tr><tr><td>13</td><td>35202-1234567-13</td><td>2025-06-02 12:53:13</td><td>2025-06-02 12:53:13</td></tr><tr><td>14</td><td>35202-1234567-14</td><td>2025-06-02 12:53:13</td><td>2025-06-02 12:53:13</td></tr><tr><td>15</td><td>35202-1234567-15</td><td>2025-06-02 12:53:13</td><td>2025-06-02 12:53:13</td></tr></tbody></table> <p>15 rows in set (0.00 sec)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Log_id              | CNIC_User           | Login_time  | Log_out_time  | 1           | 35202-1234567-1 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 | 2         | 35202-1234567-2 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 | 3             | 35202-1234567-3 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 | 4          | 35202-1234567-4 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 | 5       | 35202-1234567-5 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 | 6          | 35202-1234567-6 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 | 7        | 35202-1234567-7 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 | 8       | 35202-1234567-8 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 | 9                | 35202-1234567-9 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 | 10       | 35202-1234567-10 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 | 11          | 35202-1234567-11 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 | 12              | 35202-1234567-12 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 | 13       | 35202-1234567-13 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 | 14      | 35202-1234567-14 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 | 15         | 35202-1234567-15 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| Log_id           | CNIC_User                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Login_time          | Log_out_time        |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 1                | 35202-1234567-1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 2                | 35202-1234567-2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 3                | 35202-1234567-3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 4                | 35202-1234567-4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 5                | 35202-1234567-5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 6                | 35202-1234567-6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 7                | 35202-1234567-7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 8                | 35202-1234567-8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 9                | 35202-1234567-9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 10               | 35202-1234567-10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 11               | 35202-1234567-11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 12               | 35202-1234567-12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 13               | 35202-1234567-13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 14               | 35202-1234567-14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| 15               | 35202-1234567-15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2025-06-02 12:53:13 | 2025-06-02 12:53:13 |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| Fig 4.2.3        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                     |                     |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| Table 4          | Transaction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                     |                     |             |               |             |                 |                     |                     |           |                 |                     |                     |               |                 |                     |                     |            |                 |                     |                     |         |                 |                     |                     |            |                 |                     |                     |          |                 |                     |                     |         |                 |                     |                     |                  |                 |                     |                     |          |                  |                     |                     |             |                  |                     |                     |                 |                  |                     |                     |          |                  |                     |                     |         |                  |                     |                     |            |                  |                     |                     |          |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |
| Query            | Select * from Transaction;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          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  |        |                  |                  |         |            |            |             |                 |                  |         |                |          |       |         |                  |             |                 |            |      |                 |                  |         |            |          |             |                 |                  |         |                  |            |        |        |            |             |                 |       |       |                 |      |     |          |        |             |                 |      |       |                |        |             |        |        |             |                 |       |       |                 |      |        |            |          |             |                 |       |       |                 |        |       |           |           |             |                 |        |     |                  |      |        |            |        |             |



| Output           | <pre>mysql&gt; select * from transaction;</pre> <table><thead><tr><th>Transaction_id</th><th>Timestamp</th><th>Amount</th><th>Transaction_type</th><th>wallet_id</th></tr></thead><tbody><tr><td>1</td><td>2025-06-02 12:53:23</td><td>500.00</td><td>CT</td><td>1</td></tr><tr><td>2</td><td>2025-06-02 12:53:23</td><td>300.00</td><td>CT</td><td>2</td></tr><tr><td>3</td><td>2025-06-02 12:53:23</td><td>200.00</td><td>CT</td><td>3</td></tr><tr><td>4</td><td>2025-06-02 12:53:23</td><td>400.00</td><td>CT</td><td>4</td></tr><tr><td>5</td><td>2025-06-02 12:53:23</td><td>600.00</td><td>CT</td><td>5</td></tr><tr><td>6</td><td>2025-06-02 12:53:23</td><td>700.00</td><td>IT</td><td>6</td></tr><tr><td>7</td><td>2025-06-02 12:53:23</td><td>100.00</td><td>IT</td><td>7</td></tr><tr><td>8</td><td>2025-06-02 12:53:23</td><td>150.00</td><td>IT</td><td>8</td></tr><tr><td>9</td><td>2025-06-02 12:53:23</td><td>250.00</td><td>IN</td><td>9</td></tr><tr><td>10</td><td>2025-06-02 12:53:23</td><td>350.00</td><td>IN</td><td>10</td></tr><tr><td>11</td><td>2025-06-02 12:53:23</td><td>450.00</td><td>ML</td><td>11</td></tr><tr><td>12</td><td>2025-06-02 12:53:23</td><td>550.00</td><td>ML</td><td>12</td></tr><tr><td>13</td><td>2025-06-02 12:53:23</td><td>650.00</td><td>BP</td><td>13</td></tr><tr><td>14</td><td>2025-06-02 12:53:23</td><td>750.00</td><td>BP</td><td>14</td></tr><tr><td>15</td><td>2025-06-02 12:53:23</td><td>850.00</td><td>BP</td><td>15</td></tr></tbody></table> <pre>15 rows in set (0.00 sec)</pre> | Transaction_id   | Timestamp        | Amount    | Transaction_type | wallet_id | 1                | 2025-06-02 12:53:23 | 500.00           | CT | 1                | 2 | 2025-06-02 12:53:23 | 300.00 | CT               | 2 | 3                | 2025-06-02 12:53:23 | 200.00          | CT | 3               | 4 | 2025-06-02 12:53:23 | 400.00 | CT              | 4  | 5               | 2025-06-02 12:53:23 | 600.00          | CT | 5               | 6  | 2025-06-02 12:53:23 | 700.00 | IT | 6 | 7 | 2025-06-02 12:53:23 | 100.00 | IT | 7 | 8 | 2025-06-02 12:53:23 | 150.00 | IT | 8 | 9 | 2025-06-02 12:53:23 | 250.00 | IN | 9 | 10 | 2025-06-02 12:53:23 | 350.00 | IN | 10 | 11 | 2025-06-02 12:53:23 | 450.00 | ML | 11 | 12 | 2025-06-02 12:53:23 | 550.00 | ML | 12 | 13 | 2025-06-02 12:53:23 | 650.00 | BP | 13 | 14 | 2025-06-02 12:53:23 | 750.00 | BP | 14 | 15 | 2025-06-02 12:53:23 | 850.00 | BP | 15 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------|-----------|------------------|-----------|------------------|---------------------|------------------|----|------------------|---|---------------------|--------|------------------|---|------------------|---------------------|-----------------|----|-----------------|---|---------------------|--------|-----------------|----|-----------------|---------------------|-----------------|----|-----------------|----|---------------------|--------|----|---|---|---------------------|--------|----|---|---|---------------------|--------|----|---|---|---------------------|--------|----|---|----|---------------------|--------|----|----|----|---------------------|--------|----|----|----|---------------------|--------|----|----|----|---------------------|--------|----|----|----|---------------------|--------|----|----|----|---------------------|--------|----|----|
| Transaction_id   | Timestamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Amount           | Transaction_type | wallet_id |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 1                | 2025-06-02 12:53:23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 500.00           | CT               | 1         |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 2                | 2025-06-02 12:53:23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 300.00           | CT               | 2         |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 3                | 2025-06-02 12:53:23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 200.00           | CT               | 3         |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 4                | 2025-06-02 12:53:23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 400.00           | CT               | 4         |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 5                | 2025-06-02 12:53:23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 600.00           | CT               | 5         |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 6                | 2025-06-02 12:53:23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 700.00           | IT               | 6         |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 7                | 2025-06-02 12:53:23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 100.00           | IT               | 7         |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 8                | 2025-06-02 12:53:23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 150.00           | IT               | 8         |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 9                | 2025-06-02 12:53:23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 250.00           | IN               | 9         |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 10               | 2025-06-02 12:53:23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 350.00           | IN               | 10        |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 11               | 2025-06-02 12:53:23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 450.00           | ML               | 11        |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 12               | 2025-06-02 12:53:23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 550.00           | ML               | 12        |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 13               | 2025-06-02 12:53:23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 650.00           | BP               | 13        |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 14               | 2025-06-02 12:53:23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 750.00           | BP               | 14        |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 15               | 2025-06-02 12:53:23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 850.00           | BP               | 15        |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| Fig 4.2.4        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| Table 5          | CNIC_transfer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| Query            | Select * from CNIC_transfer;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| Output           | <pre>mysql&gt; select * from CNIC_transfer;</pre> <table><thead><tr><th>C_Transaction_id</th><th>Recipient_CNIC</th></tr></thead><tbody><tr><td>6</td><td>35202-1234567-1</td></tr><tr><td>15</td><td>35202-1234567-10</td></tr><tr><td>1</td><td>35202-1234567-11</td></tr><tr><td>2</td><td>35202-1234567-12</td></tr><tr><td>3</td><td>35202-1234567-13</td></tr><tr><td>4</td><td>35202-1234567-14</td></tr><tr><td>5</td><td>35202-1234567-15</td></tr><tr><td>7</td><td>35202-1234567-2</td></tr><tr><td>8</td><td>35202-1234567-3</td></tr><tr><td>9</td><td>35202-1234567-4</td></tr><tr><td>10</td><td>35202-1234567-5</td></tr><tr><td>11</td><td>35202-1234567-6</td></tr><tr><td>12</td><td>35202-1234567-7</td></tr><tr><td>13</td><td>35202-1234567-8</td></tr><tr><td>14</td><td>35202-1234567-9</td></tr></tbody></table> <pre>15 rows in set (0.00 sec)</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | C_Transaction_id | Recipient_CNIC   | 6         | 35202-1234567-1  | 15        | 35202-1234567-10 | 1                   | 35202-1234567-11 | 2  | 35202-1234567-12 | 3 | 35202-1234567-13    | 4      | 35202-1234567-14 | 5 | 35202-1234567-15 | 7                   | 35202-1234567-2 | 8  | 35202-1234567-3 | 9 | 35202-1234567-4     | 10     | 35202-1234567-5 | 11 | 35202-1234567-6 | 12                  | 35202-1234567-7 | 13 | 35202-1234567-8 | 14 | 35202-1234567-9     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| C_Transaction_id | Recipient_CNIC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 6                | 35202-1234567-1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 15               | 35202-1234567-10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 1                | 35202-1234567-11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 2                | 35202-1234567-12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 3                | 35202-1234567-13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 4                | 35202-1234567-14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 5                | 35202-1234567-15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 7                | 35202-1234567-2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 8                | 35202-1234567-3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 9                | 35202-1234567-4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 10               | 35202-1234567-5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 11               | 35202-1234567-6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 12               | 35202-1234567-7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 13               | 35202-1234567-8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| 14               | 35202-1234567-9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| Fig 4.2.5        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| Table 6          | Mobile Load                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |
| Query            | Select * from mobile_load;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                  |                  |           |                  |           |                  |                     |                  |    |                  |   |                     |        |                  |   |                  |                     |                 |    |                 |   |                     |        |                 |    |                 |                     |                 |    |                 |    |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |   |                     |        |    |   |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |    |                     |        |    |    |

Fig 4.2.4

Fig 4.2.5

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Output</b>  | <pre>mysql&gt; select * from mobile_load; +-----+-----+   M_Transaction_id   Mobile_number   +-----+-----+   1   03110000001     2   03110000002     3   03110000003     4   03110000004     5   03110000005     6   03110000006     7   03110000007     8   03110000008     9   03110000009     10   03110000010     11   03110000011     12   03110000012     13   03110000013     14   03110000014     15   03110000015   +-----+-----+ 15 rows in set (0.00 sec)</pre>                                                                                                                                                                                                                                  |
| Fig 4.2.6      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Table 7</b> | <b>Inter wallet transfer</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Query</b>   | Select * from interwallettransfer;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Output</b>  | <pre>mysql&gt; select * from interwallettransfer; +-----+-----+-----+   I_Transaction_id   Recipient_Phone   Recipient_Wallet_Type   +-----+-----+-----+   1   03000000001   EasyPaisa     2   03000000002   JazzCash     3   03000000003   EasyPaisa     4   03000000004   JazzCash     5   03000000005   EasyPaisa     6   03000000006   JazzCash     7   03000000007   EasyPaisa     8   03000000008   JazzCash     9   03000000009   EasyPaisa     10   03000000010   JazzCash     11   03000000011   EasyPaisa     12   03000000012   JazzCash     13   03000000013   EasyPaisa     14   03000000014   JazzCash     15   03000000015   EasyPaisa   +-----+-----+-----+ 15 rows in set (0.00 sec)</pre> |
| Fig 4.2.7      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Table 8</b> | <b>Internal Transfer</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Query</b>   | Select * from internal_transfer;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

Output

```
mysql> select * from internaltransfer;
```

| IN_Transaction_id | Recipient_Phone |
|-------------------|-----------------|
| 10                | 03001000000     |
| 11                | 03001100000     |
| 1                 | 03001111111     |
| 12                | 03001200000     |
| 13                | 03001300000     |
| 14                | 03001400000     |
| 15                | 03001500000     |
| 2                 | 03002222222     |
| 3                 | 03003333333     |
| 4                 | 03004444444     |
| 5                 | 03005555555     |
| 6                 | 03006666666     |
| 7                 | 03007777777     |
| 8                 | 03008888888     |
| 9                 | 03009999999     |

```
15 rows in set (0.00 sec)
```

Fig 4.2.8

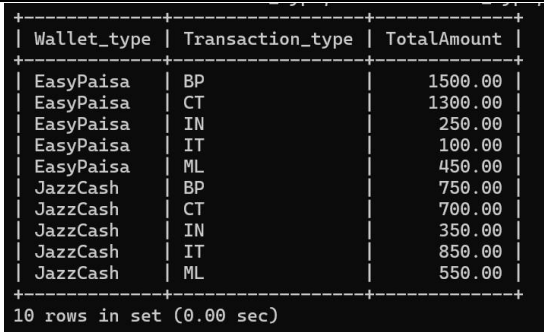
### 4.3. QUERIES RESULTS:

|               |                                                                                                                    |
|---------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Task 1</b> | Get all users along with their wallet balance and wallet type.                                                     |
| <b>Query</b>  | SELECT u.Name, u.CNIC_user, w.Balance, w.Wallet_type<br>FROM User u<br>JOIN Wallet w ON u.CNIC_user = w.CNIC_user; |

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Output</b> | <pre> +-----+-----+-----+-----+   Name   CNIC_user   Balance   Wallet_type   +-----+-----+-----+-----+   Ali   35202-1234567-1   9500.00   EasyPaisa     Nadia   35202-1234567-10   650.00   JazzCash     Rehan   35202-1234567-11   9050.00   EasyPaisa     Tania   35202-1234567-12   7950.00   JazzCash     Imran   35202-1234567-13   6850.00   EasyPaisa     Zara   35202-1234567-14   5750.00   JazzCash     Kamran   35202-1234567-15   4650.00   EasyPaisa     Sara   35202-1234567-2   8700.00   JazzCash     Usman   35202-1234567-3   7800.00   EasyPaisa     Ayesha   35202-1234567-4   6600.00   JazzCash     Bilal   35202-1234567-5   5400.00   EasyPaisa     Hina   35202-1234567-6   4300.00   JazzCash     Hamza   35202-1234567-7   3900.00   EasyPaisa     Farah   35202-1234567-8   2850.00   JazzCash     Danish   35202-1234567-9   1750.00   EasyPaisa   +-----+-----+-----+-----+ 15 rows in set (0.00 sec) </pre> <p>Fig 4.3.1</p>                                                                                                                                                               |
| <b>Task 2</b> | List all transactions along with user info and wallet type.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Query</b>  | <pre> SELECT t.Transaction_id, t.Timestamp, t.Amount, u.Name, w.Wallet_type FROM Transaction t JOIN Wallet w ON t.wallet_id = w.Wallet_id JOIN User u ON w.CNIC_user = u.CNIC_user; </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Output</b> | <pre> +-----+-----+-----+-----+-----+-----+   Transaction_id   Timestamp   Amount   Name   Wallet_type   +-----+-----+-----+-----+-----+-----+   1   2025-06-02 12:53:23   500.00   Ali   EasyPaisa     2   2025-06-02 12:53:23   300.00   Sara   JazzCash     3   2025-06-02 12:53:23   200.00   Usman   EasyPaisa     4   2025-06-02 12:53:23   400.00   Ayesha   JazzCash     5   2025-06-02 12:53:23   600.00   Bilal   EasyPaisa     6   2025-06-02 12:53:23   700.00   Hina   JazzCash     7   2025-06-02 12:53:23   100.00   Hamza   EasyPaisa     8   2025-06-02 12:53:23   150.00   Farah   JazzCash     9   2025-06-02 12:53:23   250.00   Danish   EasyPaisa     10   2025-06-02 12:53:23   350.00   Nadia   JazzCash     11   2025-06-02 12:53:23   450.00   Rehan   EasyPaisa     12   2025-06-02 12:53:23   550.00   Tania   JazzCash     13   2025-06-02 12:53:23   650.00   Imran   EasyPaisa     14   2025-06-02 12:53:23   750.00   Zara   JazzCash     15   2025-06-02 12:53:23   850.00   Kamran   EasyPaisa   +-----+-----+-----+-----+-----+-----+ 15 rows in set (0.00 sec) </pre> <p>Fig 4.3.2</p> |
| <b>Task 3</b> | Count total transactions per wallet type, showing only those with more than 3 transactions, sorted by count descending.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Query</b>  | <pre> SELECT w.Wallet_type, COUNT(t.Transaction_id) AS TotalTransactions FROM Wallet w JOIN Transaction t ON w.Wallet_id = t.wallet_id GROUP BY w.Wallet_type HAVING COUNT(t.Transaction_id) &gt; 3 ORDER BY TotalTransactions DESC; </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Output</b> | <pre> +-----+-----+   Wallet_type   TotalTransactions   +-----+-----+   EasyPaisa   8     JazzCash   7   +-----+-----+ 2 rows in set (0.00 sec) </pre> <p>Fig 4.3.3</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|               |                                                                                 |
|---------------|---------------------------------------------------------------------------------|
| <b>Task 4</b> | Sum of amounts by transaction type for each wallet type, sorted by wallet type. |
|---------------|---------------------------------------------------------------------------------|

|              |                                                                                                                                                                                                                                        |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Query</b> | <pre>SELECT w.Wallet_type, t.Transaction_type, SUM(t.Amount) AS TotalAmount FROM Wallet w JOIN Transaction t ON w.Wallet_id = t.wallet_id GROUP BY w.Wallet_type, t.Transaction_type ORDER BY w.Wallet_type, t.Transaction_type;</pre> |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Output</b> |  <pre> +-----+-----+-----+   Wallet_type   Transaction_type   TotalAmount   +-----+-----+-----+   EasyPaisa     BP                 1500.00         EasyPaisa     CT                 1300.00         EasyPaisa     IN                 250.00          EasyPaisa     IT                 100.00          EasyPaisa     ML                 450.00          JazzCash      BP                 750.00          JazzCash      CT                 700.00          JazzCash      IN                 350.00          JazzCash      IT                 850.00          JazzCash      ML                 550.00        +-----+-----+-----+ 10 rows in set (0.00 sec) </pre> <p>Fig 4.3.4</p> |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|               |                                                 |
|---------------|-------------------------------------------------|
| <b>Task 5</b> | Find users who have never done any transaction. |
|---------------|-------------------------------------------------|

|              |                                                                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Query</b> | <pre>SELECT Name, CNIC_user FROM User WHERE CNIC_user NOT IN (     SELECT w.CNIC_user     FROM Wallet w     JOIN Transaction t ON w.Wallet_id = t.wallet_id );</pre> |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|               |                                                                                                                                                  |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Output</b> |  <pre> -&gt; ); Empty set (0.00 sec) </pre> <p>Fig 4.3.5</p> |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------|

|               |                                                                          |
|---------------|--------------------------------------------------------------------------|
| <b>Task 6</b> | Get wallets with balance higher than the average balance of all wallets. |
|---------------|--------------------------------------------------------------------------|

|              |                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Query</b> | <pre>SELECT Wallet_id, CNIC_user, Balance FROM Wallet WHERE Balance &gt; (SELECT AVG(Balance) FROM Wallet);</pre> |
|--------------|-------------------------------------------------------------------------------------------------------------------|

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| <b>Output</b> | <pre> +-----+-----+-----+   Wallet_id   CNIC_user        Balance   +-----+-----+-----+           14   35202-1234567-14   5750.00              4   35202-1234567-4    6600.00             13   35202-1234567-13   6850.00              3   35202-1234567-3    7800.00             12   35202-1234567-12   7950.00              2   35202-1234567-2    8700.00             11   35202-1234567-11   9050.00              1   35202-1234567-1    9500.00   +-----+-----+-----+ 8 rows in set (0.00 sec) </pre> <p>Fig 4.3.6</p>                                                                                                                                                                                                                                                         |  |  |
| <b>Task 7</b> | Find all transactions where the amount is at least the average amount for that wallet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  |
| <b>Query</b>  | <pre> SELECT Transaction_id, wallet_id, Amount FROM Transaction t1 WHERE Amount &gt;= (     SELECT AVG(Amount)     FROM Transaction t2     WHERE t2.wallet_id = t1.wallet_id ); </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  |
| <b>Output</b> | <pre> +-----+-----+-----+   Transaction_id   wallet_id   Amount   +-----+-----+-----+                1           1   500.00                  2           2   300.00                  3           3   200.00                  4           4   400.00                  5           5   600.00                  6           6   700.00                  7           7   100.00                  8           8   150.00                  9           9   250.00                 10          10   350.00                 11          11   450.00                 12          12   550.00                 13          13   650.00                 14          14   750.00                 15          15   850.00   +-----+-----+-----+ 15 rows in set (0.00 sec) </pre> <p>Fig 4.3.7</p> |  |  |
| <b>Task 8</b> | List all users who logged in today along with their latest login time.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  |
| <b>Query</b>  | <pre> SELECT u.Name, MAX(l.Login_time) AS Last_Login FROM User u JOIN Logs l ON u.CNIC_user = l.CNIC_User WHERE DATE(l.Login_time) = CURDATE() GROUP BY u.Name; </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  |

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Output</b>  | <pre> +-----+-----+   Name     Last_Login   +-----+-----+   Ali       2025-06-02 12:53:13     Nadia     2025-06-02 12:53:13     Rehan     2025-06-02 12:53:13     Tania     2025-06-02 12:53:13     Imran     2025-06-02 12:53:13     Zara      2025-06-02 12:53:13     Kamran    2025-06-02 12:53:13     Sara      2025-06-02 12:53:13     Usman     2025-06-02 12:53:13     Ayesha    2025-06-02 12:53:13     Bilal     2025-06-02 12:53:13     Hina      2025-06-02 12:53:13     Hamza     2025-06-02 12:53:13     Farah     2025-06-02 12:53:13     Danish    2025-06-02 12:53:13   +-----+-----+ 15 rows in set (0.00 sec) </pre> <p>Fig 4.3.8</p> |
| <b>Task 9</b>  | List all bill payments grouped by bill type with total amount paid, only bill types with total payment over 1000.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Query</b>   | <pre> SELECT b.Bill_type, SUM(t.Amount) AS TotalPaid FROM Bill_payment b JOIN Transaction t ON b.B_Transaction_id = t.Transaction_id GROUP BY b.Bill_type HAVING SUM(t.Amount) &gt; 1000 ORDER BY TotalPaid DESC; </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Output</b>  | <pre> +-----+-----+   Bill_type   TotalPaid   +-----+-----+   Telephone   1800.00       Electricity   1650.00       Internet    1400.00     +-----+-----+ 3 rows in set (0.00 sec) </pre> <p>Fig 4.3.9</p>                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Task 10</b> | Retrieve user names and their total transaction count ordered by count descending.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Query</b>   | <pre> SELECT u.Name, COUNT(t.Transaction_id) AS TransactionCount FROM User u JOIN Wallet w ON u.CNIC_user = w.CNIC_user JOIN Transaction t ON w.Wallet_id = t.wallet_id GROUP BY u.Name ORDER BY TransactionCount DESC; </pre>                                                                                                                                                                                                                                                                                                                                                                                                                          |

|               |                   |
|---------------|-------------------|
| <b>Output</b> | <p>Fig 4.3.10</p> |
|---------------|-------------------|

## Chapter 05 : Interface Design

### 5.1 LANGUAGE/Framework

The GUI client for the ewallet system was developed using **Python** with the **Tkinter** framework. Tkinter is a built-in GUI library in Python that provides a fast and easy way to build interactive interfaces. It was selected due to the following reasons:

- **Ease of Use:** Tkinter allows for rapid GUI development without the need for extensive boilerplate code.
- **Customizability:** It supports full customization of UI elements including fonts, colors, styles, and layouts.
- **Cross-platform Support:** Tkinter applications are portable across Windows, macOS, and Linux.
- **Integration with mysql:** Python supports mysql connectivity with the mysql-connector-python library, enabling seamless database operations within the GUI.

#### Key features used in the GUI:

- Interactive frames for actions such as viewing, adding, updating, and deleting data.



- Beautifully styled welcome screen with emojis, gradient effects, and user-friendly labels.
- Dynamic dropdowns populated from the database (e.g., table and column names).
- Real-time data validation and feedback using messagebox.

## 5.2 DATABASE CONNECTIVITY:

The GUI connects to a **MySQL** database named **walletsystem** using the **mysql.connector** library. The connection is established

```
Code
import mysql.connector

def connect_to_database():
    return mysql.connector.connect(
        host="localhost",
        user="root",
        password="zaira123786",
        database="walletsystem"
    )
```

Error handling is implemented for actions like search, insert, update, and delete using try-except blocks. For example, duplicate entries are caught and presented to the user with custom error messages.

## 5.3. STORED PROCEDURES AND FUNCTIONS:

In the EWallet system, several **stored functions and procedures** were implemented to handle database operations more efficiently and cleanly. These scripts help keep the backend logic organized, reusable, and secure. They also simplify interaction between the Python GUI and the database by encapsulating common operations.

### 5.3.1. Stored Functions

#### 1.GetWalletBalance

```
CREATE FUNCTION GetWalletBalance(userCNIC VARCHAR(15))
RETURNS DECIMAL(10,2)
DETERMINISTIC
READS SQL DATA
BEGIN
    DECLARE balance DECIMAL(10,2);
    SELECT w.Balance INTO balance
    FROM Wallet w
    WHERE w.CNIC_user = userCNIC;
    RETURN balance;
END //
```

**Purpose:** This function returns the wallet balance for a given user CNIC.

**Use in GUI:** Helps display the current wallet balance of a user directly in the application without writing the query repeatedly.

**Enhancement:** Speeds up balance checking and reduces code repetition in the GUI logic.

## 2. GetUserFullName

```
CREATE FUNCTION GetUserFullName(userCNIC VARCHAR(15))
RETURNS VARCHAR(100)
DETERMINISTIC
READS SQL DATA
BEGIN
    DECLARE fullName VARCHAR(100);
    SELECT CONCAT(Name, ' ', Father_Name) INTO fullName
    FROM User
    WHERE CNIC_user = userCNIC;
    RETURN fullName;
END //
```

**Purpose:** Returns the full name of a user (Name + Father Name) based on CNIC.

**Use in GUI:** This is useful when showing personalized greetings or generating reports.

**Enhancement:** Increases user-friendliness by displaying complete identity instead of only CNIC.

## 3. CountUserTransactions

```
CREATE FUNCTION CountUserTransactions(userCNIC VARCHAR(15))
RETURNS INT
DETERMINISTIC
READS SQL DATA
BEGIN
    DECLARE countTrans INT;
    SELECT COUNT(*) INTO countTrans
    FROM Wallet w
```

```
JOIN Transaction t ON w.Wallet_id = t.wallet_id

WHERE w.CNIC_user = userCNIC;

RETURN countTrans;

END //
```

**Purpose:** Counts the number of transactions made by a user.

**Use in GUI:** Shows transaction activity per user to track usage or display statistics.

**Enhancement:** Supports user activity analysis directly from the application.

#### 4. GetLastLoginTime

```
CREATE FUNCTION GetLastLoginTime(userCNIC VARCHAR(15))

RETURNS DATETIME

DETERMINISTIC

READS SQL DATA

BEGIN

    DECLARE lastLogin DATETIME;

    SELECT MAX(Login_time) INTO lastLogin

    FROM Logs

    WHERE CNIC_User = userCNIC;

    RETURN lastLogin;

END //
```

**Purpose:** Returns the most recent login time for a user.

**Use in GUI:** Displays the user's last login time for security or audit purposes.

**Enhancement:** Builds transparency and trust with users by showing last activity.

### 5.3.2. Stored Procedures

#### 1. AddNewUser

```
CREATE PROCEDURE AddNewUser(  
    IN p_CNIC VARCHAR(15),  
    IN p_Name VARCHAR(50),  
    IN p_FatherName VARCHAR(50),  
    IN p_Email VARCHAR(50),  
    IN p_Gender VARCHAR(10),  
    IN p_Province VARCHAR(30),  
    IN p_District VARCHAR(30),  
    IN p_Tehsil VARCHAR(30),  
    IN p_Phone VARCHAR(15)  
)  
  
BEGIN  
  
    INSERT INTO User (CNIC_user, Name, Father_Name, Email, Gender, Province, District,  
    Tehsil, Phone_num)  
  
    VALUES (p_CNIC, p_Name, p_FatherName, p_Email, p_Gender, p_Province, p_District,  
    p_Tehsil, p_Phone);  
  
END //
```

**Purpose:** Adds a new user to the database.

**Use in GUI:** Called when the user fills the "Add User" form in the GUI.

**Enhancement:** Keeps user registration logic centralized and consistent.

#### 2. UpdateWalletBalance

```
CREATE PROCEDURE UpdateWalletBalance(  

```

```
    IN p_WalletId INT,  
    IN p_NewBalance DECIMAL(10,2)  
)  
BEGIN  
    UPDATE Wallet  
    SET Balance = p_NewBalance  
    WHERE Wallet_id = p_WalletId;  
END //
```

**Purpose:** Updates the balance of a specific wallet.

**Use in GUI:** Used after deposits or withdrawals.

**Enhancement:** Ensures balance changes are secure and trackable via procedure calls.

### 3. LogUserLogin

```
CREATE PROCEDURE LogUserLogin(  
    IN p_CNIC VARCHAR(15)  
)  
BEGIN  
    INSERT INTO Logs (CNIC_User, Login_time)  
    VALUES (p_CNIC, NOW());  
END //
```

**Purpose:** Logs each time a user logs into the system.

**Use in GUI:** Automatically called when a user starts a session or logs in.

**Enhancement:** Supports login tracking for security and monitoring.

### 4. RecordTransaction

```

CREATE PROCEDURE RecordTransaction(

    IN p_TransactionId INT,

    IN p_Timestamp DATETIME,

    IN p_Amount DECIMAL(10,2),

    IN p_Type CHAR(2),

    IN p_wallet_id INT

)

BEGIN

    INSERT INTO Transaction (Transaction_id, Timestamp, Amount, Transaction_type, wallet_id)

    VALUES (p_TransactionId, p_Timestamp, p_Amount, p_Type, p_wallet_id);

END //

```

**Purpose:** Adds a new transaction record to the database.

**Use in GUI:** Called when a user performs a transaction (e.g., deposit or withdrawal).

**Enhancement:** Makes transaction logging accurate and automated from the application.

## 5.4. INTERFACES:

### Welcome Screen

Features a modern card-style interface with emojis, a gradient design, and a “Get Started” button. It introduces the app as a secure and stylish wallet system.

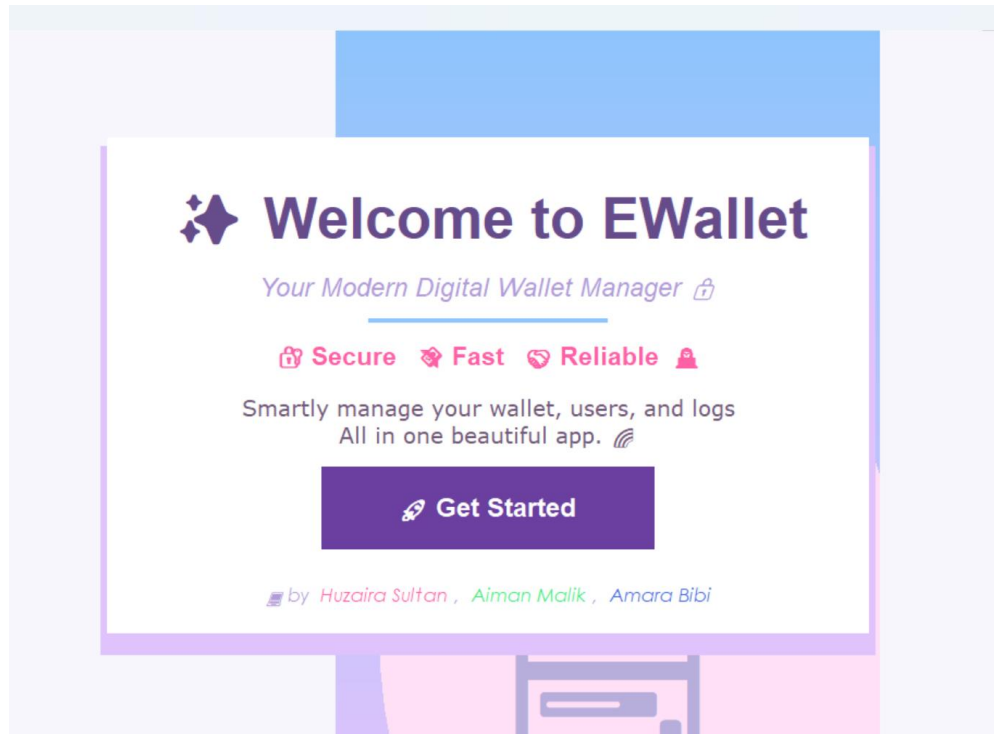


Fig 1.1

## Main Menu

Provides buttons for:

- Viewing all tables
- Viewing table data
- Adding, deleting, updating, or searching records

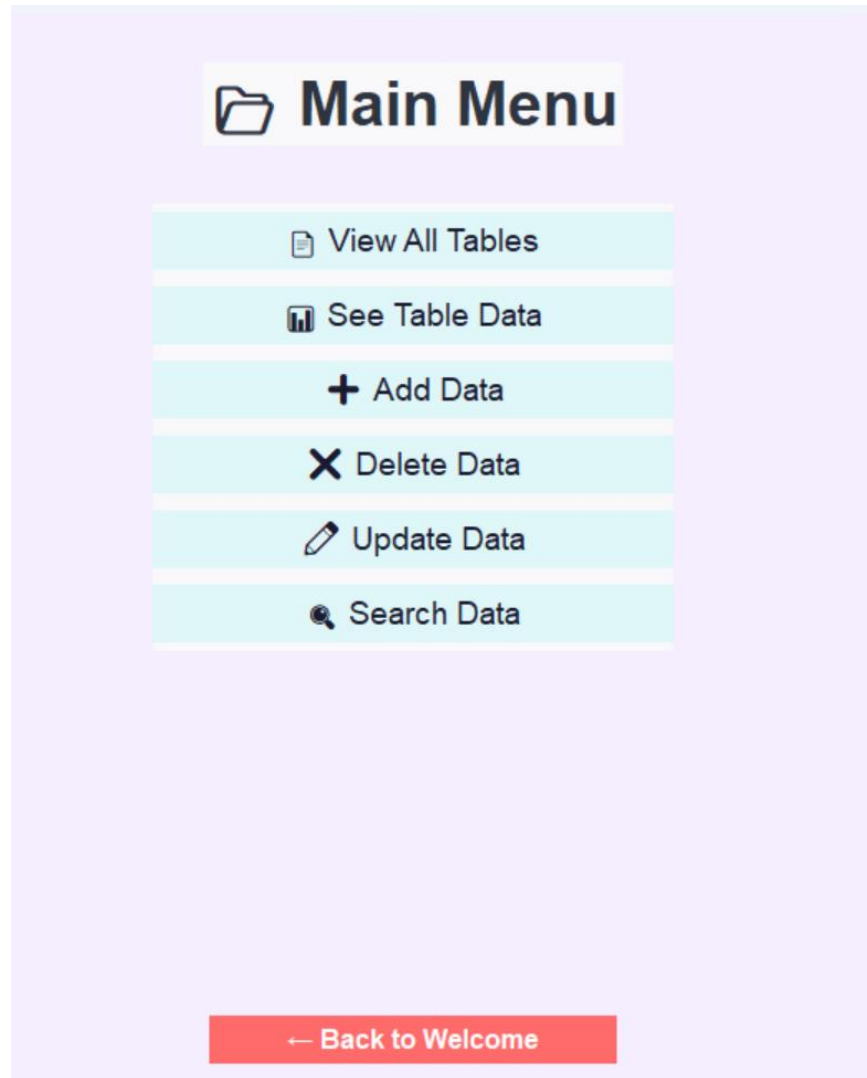


Fig 1.2

### Table View Interface

Allows the user to select any table from the database and view its contents in a structured, scrollable Treeview table.



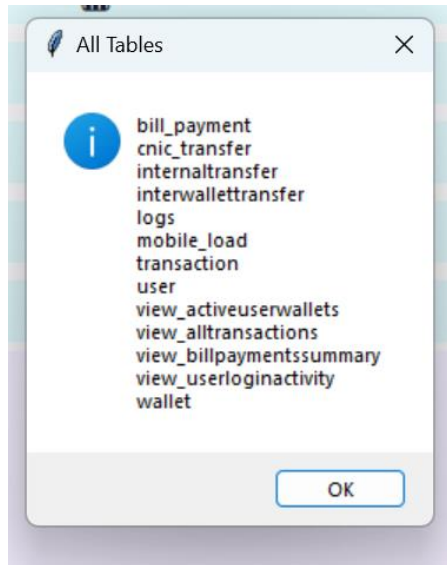


Fig 1.3

### Data Entry form

Dynamically generated based on the table schema. Users can add or update data through clean entry fields.

CNIC\_user

38302-7890789-2

Name

huzaira

Father\_Name

sultan

Email

zaira@gmail.com

Gender

female

Province

punjab

District

mianwali

Tehsil

minwali

Phone\_num

03004509876

password

huz123

Submit

Fig 1.4

### Search/Delete Actions

Dropdowns allow users to pick columns to filter by. Based on input, the app displays or deletes the matching records.

Each interface supports:

- Error messages using messagebox
- Validation (e.g., CNIC must be 13 digits)
- Responsiveness and intuitive navigation

Select Column: Transaction\_id

Enter Value:

3

Submit

Back

Fig 1.5

Search Results

| Transaction_id | Timestamp        | Amount | Transaction_type | wallet_id |
|----------------|------------------|--------|------------------|-----------|
| 3              | 2025-06-02 12:53 | 200.00 | BP               | 3         |

Select Column: Transaction\_id

Enter Value:

3

Submit

Back

## **6. CONCLUSION**

### **6.1. LESSONS LEARNED:**

During this project, we learned many valuable things. Technically, we improved our skills in Python programming, GUI development using Tkinter, and how to connect and work with mysql databases. We also learned how to handle errors and validate user input effectively. From a project management point of view, we learned how to divide tasks, manage time, and meet deadlines. Teamwork played an important role, and we saw how sharing ideas and helping each other can lead to better results. Communication and planning were key to staying organized.

### **6.2. CHALLENGES AND SOLUTIONS:**

One of the biggest challenges was connecting the GUI with the mysql database. We faced errors with user inputs and duplicate data. We solved these by using proper error handling and adding checks for valid input (like CNIC format). Designing the GUI to look clean and user-friendly also took time. We solved this by testing different layouts and getting feedback from each other. Sometimes, the data wasn't saving or updating correctly, which we fixed by carefully checking our SQL queries and using debug messages.

### **6.3. FUTURE WORK AND IMPROVEMENTS:**

In the future, we can improve the system by adding user login and authentication, so only registered users can access the wallet. We could also add graphs or charts to show wallet activity visually. Another improvement would be to create a mobile version or web-based version of the ewallet. Adding email or SMS notifications for transactions would also be useful.

### **6.4. FINAL THOUGHTS:**

This project helped us grow as students and developers. We turned an idea into a working application, and that gave us confidence. It also showed us the importance of testing and clear code. Overall, it was a great learning experience, and we are proud of what we achieved.

---

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1

]

[ "lucidchart," [Online]. Available: [https://lucid.app/lucidchart/1d53e44b-8bed-4057-ace1-](https://lucid.app/lucidchart/1d53e44b-8bed-4057-ace1-21c6050c986a2/edit?viewport_loc=-3339%2C-)

[21c6050c986a2/edit?viewport\\_loc=-3339%2C-](https://lucid.app/lucidchart/1d53e44b-8bed-4057-ace1-21c6050c986a2/edit?viewport_loc=-3339%2C-5341%2C7494%2C11766%2C0_0&invitationId=inv_06cc6efc-a4d3-4c5e-b306-0ff1dff1fb0)

[ "chatgpt," [Online]. Available:

[3https://chatgpt.com/?utm\\_source=google&utm\\_medium=paidsearch\\_brand&utm\\_campaign=](https://chatgpt.com/?utm_source=google&utm_medium=paidsearch_brand&utm_campaign=)

[\] DEPT\\_SEM\\_Google\\_Brand\\_Acquisition\\_EMEA\\_France\\_Consumer\\_CPA\\_BAU\\_Mix\\_English  
&utm\\_term=chatgpt&gad\\_source=1&gad\\_campaignid=22443202931&gbraid=0AAAAAIW-  
UUDh2ka6WA-bbrV2qrad78k7&g.](https://chatgpt.com/?utm_source=google&utm_medium=paidsearch_brand&utm_campaign=DEPT_SEM_Google_Brand_Acquisition_EMEA_France_Consumer_CPA_BAU_Mix_English&utm_term=chatgpt&gad_source=1&gad_campaignid=22443202931&gbraid=0AAAAAIW-UUDh2ka6WA-bbrV2qrad78k7&g)